AI 101 for Mechanical Engineers · Module 16

Applications with different evidence needs

Lesson 16.1

What you will learn

> Explain why applications need different evidence, how a surrogate supports design search, how an objective differs from constraints and feasibility, and which data connection defines a digital twin.

Lesson overview

Module 16 · Lesson 1

Applications with different evidence needs

AI 101 for Mechanical Engineers

1 Surrogates speed up search, not sign-off

Expensive simulation or experiment → Surrogate screens many candidates → Re-evaluate promising designs

Verification: was the model or calculation implemented correctly?

Physical validation: is it adequate against evidence from the real system?

2 Objective and constraints are different questions

Objective	**Constraints**
e.g. reduce pump power	e.g. minimum flow, maximum stress

A design with low predicted power that breaks a flow or stress limit is **infeasible**. Check each constraint and its units separately.

3 Digital model, shadow or twin? Follow the data

LEVEL	AUTOMATIC DATA FLOW	EQUIPMENT ⇄ MODEL
Digital model	None; data are exchanged manually. An isolated CAD model is an example.	
Digital shadow	Equipment → model, e.g. live pressure, flow or temperature for monitoring.	
Digital twin	Both directions: the model also sends updates, such as settings, back to the equipment.	

Equipment on the left, digital representation on the right. Classification after Kritzinger et al.; definitions vary, so state the one your project uses.

4 Physics-informed learning

$$\text{Loss} = \text{data error} + \text{weighted equation residual}$$

A prediction that breaks a governing equation or conservation law, such as an energy balance, is penalized during training.

A **PINN** is one approach: a neural network trained with governing-equation residuals and boundary or initial conditions in its loss. It is not one fixed architecture.

Key takeaway

Match the evidence to the engineering decision.

Surrogates screen, constraints decide feasibility, data connections define a twin, and physics can shape the loss.

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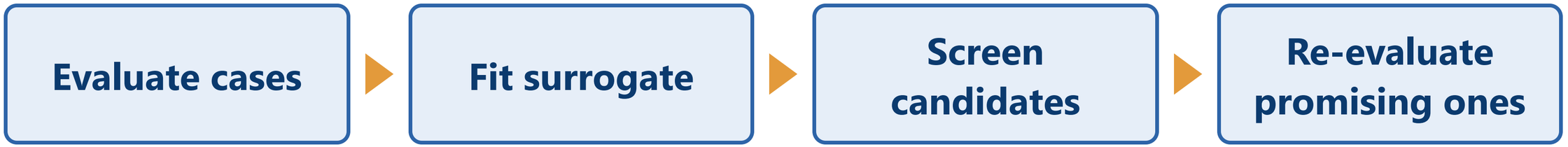
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The decision determines the model and evidence

- AI can support design, manufacturing, thermofluids, maintenance and robotics
- Screening designs, thermal estimates, fault finding and robot coordination differ
- Example: motor-driven centrifugal pump with piped connections and instruments
- State the decision, available inputs and intended operating conditions

> A strong score on one dataset does not by itself show suitability elsewhere.

Surrogates accelerate search, not sign-off



- High-fidelity simulation can be expensive for every design [20]
- Accuracy supported only where evidence represents the conditions
- Does not replace verification or validation, or authorize hardware claims

Bayesian optimization prioritizes; it does not prove

- Uncertainty-aware search can use a surrogate to choose candidates [84]
- It can make expensive searches more efficient
- It does not verify the implementation or prove constraint satisfaction
- It does not establish physical validity

Objective and constraints answer different questions

Objective

- States the performance goal [19]
- e.g. reduce a pump’s electrical power demand

Constraints and bounds

- Define permitted designs or outcomes
- e.g. minimum required flow, maximum allowable stress

> The feasible design space contains only candidates that satisfy all stated constraints.

A good objective can still be infeasible

- Low predicted power but a violated stress or flow limit
- Such a design is not acceptable
- Check constraint values and units separately from the objective

The data connection defines a digital twin

Level	Data exchange with equipment
Digital model (e.g. isolated CAD)	Manual only
Digital shadow	Automatic from equipment; no automatic action back
Digital twin	Automatic in both directions

Definitions vary [86], [95]. This widely used manufacturing classification follows [85], [95].

Live data links bring new responsibilities

- The data connection, not visual similarity, is the distinction
- State which definition and data connections your project uses
- Active sensor links create data-integrity and cybersecurity responsibilities
- Investigate model–reading disagreement before acting on a prediction

A PINN is one physics-informed approach

Physics-informed learning

- Builds physical equations or constraints into learning [87], [88]
- e.g. penalty for violating a governing equation

PINN

- Loss includes governing-equation residual and boundary/initial conditions [87]
- No single required network architecture

PINN is not a synonym for every hybrid model. Physical structure: e.g. F = ma, thermodynamic constraints.

CHECK YOUR UNDERSTANDING

A surrogate predicts a promising pump design. What must happen before the design is accepted?

Answer before moving on.

CHECK YOUR UNDERSTANDING

A surrogate predicts a promising pump design. What must happen before the design is accepted?

> Re-evaluate it with the original simulation or experiment, check every constraint and its units separately from the objective, and validate it against real-system evidence.

CHECK YOUR UNDERSTANDING

What makes a digital twin different from an isolated CAD model?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What makes a digital twin different from an isolated CAD model?

An automatic data connection. In this classification, automatic data from the equipment makes a digital shadow; a twin also sends data back automatically. Visual similarity is not the distinction.

Key takeaways

- Match the model and the evidence to the engineering decision
- Surrogates speed search; re-evaluate promising candidates with simulation or experiment
- The objective states the goal; constraints decide feasibility
- Common classification: shadow gets automatic data; twin both ways; CAD neither
- A PINN is one physics-informed approach, not a single architecture

References (1/2)

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Numbers match the course reference list. The written lesson is the authoritative text.

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